4	(a)	waves pass through the elements / gaps / slits in the grating	M1	
		spread into geometric shadow	A1	[2]
	(b) (i) 1.	displacements add to give resultant displacement	B1	
		each wavelength travels the same path difference or are in phase	B1	
		hence produce a maximum	A0	[2]
	2.	to obtain a maximum the path difference must be λ or phase difference 360° / 2π rad	B1	
		λ of red and blue are different	B1	
		hence maxima at different angles / positions	A0	[2]
	(ii)	$n\lambda = d \sin\theta$	C1	
		$N = \sin 61° / (2 \times 625 \times 10^{-9}) = 7.0 \times 10^5$	A1	[2]
	(iii)	$n\lambda = 2 \times 625$ is a constant (1250)	C1	
		$n = 1 \rightarrow \lambda = 1250$ outside visible		
		$n = 3 \rightarrow \lambda = 417$ in visible		
		$n = 4 \rightarrow \lambda = 312.5$ outside visible		
		$\lambda = 420$ nm	A1	[2]
5	(a)	when the load is removed then the wire / body object does not return to its original shape /		